

Roadmap for Two-Field False Vacuum Decay Using the Green-Function Method

1 Introduction

This document outlines the full workflow for computing the false-vacuum decay rate in a two-scalar-field theory using the Green-function method for one-loop corrections. The steps follow the logical progression familiar from the one-field case, but are generalised to a multi-field system.

We consider a Lagrangian of the form

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\phi_1)^2 + \frac{1}{2}(\partial_\mu\phi_2)^2 - V(\phi_1, \phi_2), \quad (1)$$

where $V(\phi_1, \phi_2)$ is a renormalisable potential chosen to admit a metastable vacuum:

$$V(\phi_1, \phi_2) = -\frac{1}{2}\mu_1^2\phi_1^2 + \frac{\lambda_1}{4}\phi_1^4 - \frac{1}{2}\mu_2^2\phi_2^2 + \frac{\lambda_2}{4}\phi_2^4 + \lambda_{12}\phi_1^2\phi_2^2 + \kappa\phi_1^3 + \dots \quad (2)$$

The ellipsis indicates that additional cubic or linear terms can be added to lift degeneracies as needed.

2 1. Vacuum Structure and Minima

2.1 1.1 Stationary Point Equations

The starting point is to determine all extrema of the potential:

$$\frac{\partial V}{\partial\phi_1} = 0, \quad \frac{\partial V}{\partial\phi_2} = 0. \quad (3)$$

In practice, these equations are solved numerically for given parameter values.

2.2 1.2 Hessian and Classification

At each stationary point (ϕ_1^*, ϕ_2^*) , compute the Hessian

$$H_{ij}(\phi^*) = \left. \frac{\partial^2 V}{\partial\phi_i\partial\phi_j} \right|_{\phi=\phi^*}, \quad i, j = 1, 2. \quad (4)$$

Diagonalising H_{ij} gives eigenvalues $\lambda_{H,1}, \lambda_{H,2}$:

- Both $\lambda_{H,i} > 0$: local minimum (candidate vacuum).
- One $\lambda_{H,i} < 0$: saddle point.
- Both $\lambda_{H,i} < 0$: local maximum.

2.3 1.3 False and True Vacua

Among the local minima, identify:

- The *false vacuum* (FV), $(\phi_{1,\text{FV}}, \phi_{2,\text{FV}})$, with higher vacuum energy.
- The *true vacuum* (TV), $(\phi_{1,\text{TV}}, \phi_{2,\text{TV}})$, with lower vacuum energy.

These will serve as boundary conditions for the bounce solution.

3 2. Bounce Solution and Classical Action

3.1 2.1 Euclidean Action

After Wick rotation to Euclidean time, the Euclidean action reads

$$S_E[\phi_1, \phi_2] = \int d^4x \left[\frac{1}{2}(\partial_\mu \phi_1)^2 + \frac{1}{2}(\partial_\mu \phi_2)^2 + V(\phi_1, \phi_2) \right]. \quad (5)$$

Assuming $O(4)$ symmetry, the fields depend only on the radius $r = \sqrt{\tau^2 + \mathbf{x}^2}$:

$$\phi_i(x) = \phi_i(r), \quad i = 1, 2. \quad (6)$$

Then

$$S_E = 2\pi^2 \int_0^\infty dr r^3 \left[\frac{1}{2} \left(\frac{d\phi_1}{dr} \right)^2 + \frac{1}{2} \left(\frac{d\phi_2}{dr} \right)^2 + V(\phi_1, \phi_2) \right]. \quad (7)$$

3.2 2.2 Bounce Equations of Motion

The bounce is obtained by extremising S_E , leading to the coupled ODEs

$$\phi_1''(r) + \frac{3}{r}\phi_1'(r) = \frac{\partial V}{\partial \phi_1}(\phi_1(r), \phi_2(r)), \quad (8)$$

$$\phi_2''(r) + \frac{3}{r}\phi_2'(r) = \frac{\partial V}{\partial \phi_2}(\phi_1(r), \phi_2(r)), \quad (9)$$

with boundary conditions

$$\phi_i'(0) = 0, \quad (10)$$

$$\phi_i(r \rightarrow \infty) = \phi_{i,\text{FV}}, \quad i = 1, 2. \quad (11)$$

These are solved numerically (e.g. shooting, relaxation, or path deformation) to obtain the bounce profiles $\phi_{i,b}(r)$.

3.3 2.3 Classical Bounce Action

The classical tunnelling exponent is given by the difference of Euclidean actions:

$$B = S_E[\phi_{1,b}, \phi_{2,b}] - S_E[\phi_{1,\text{FV}}, \phi_{2,\text{FV}}]. \quad (12)$$

At tree level, the leading contribution to the decay rate per unit volume is

$$\frac{\Gamma}{V} \sim A_{\text{tree}} e^{-B}, \quad (13)$$

where A_{tree} is a prefactor to be refined by the one-loop calculation.

4 3. Quadratic Fluctuations and Green Functions

4.1 3.1 Quadratic Fluctuation Operator

Expand the fields around the bounce:

$$\phi_i(x) = \phi_{i,b}(x) + \eta_i(x), \quad i = 1, 2. \quad (14)$$

The Euclidean action to quadratic order in fluctuations is

$$S_E^{(2)} = \frac{1}{2} \int d^4x \eta_i(x) \mathcal{M}_{ij} \eta_j(x), \quad (15)$$

with the fluctuation operator

$$\mathcal{M}_{ij} = -\delta_{ij} \partial^2 + V_{ij}(\phi_b), \quad V_{ij}(\phi_b) = \left. \frac{\partial^2 V}{\partial \phi_i \partial \phi_j} \right|_{\phi=\phi_b(x)}. \quad (16)$$

This is a 2×2 matrix of differential operators in coordinate space.

4.2 3.2 Green-Function Equation

Define the matrix Green function $G_{ij}(x, x')$ by

$$\mathcal{M}_{ik} G_{kj}(x, x') = \delta_{ij} \delta^{(4)}(x - x'). \quad (17)$$

The Green function encodes the spectrum of fluctuations and is the key object for computing the functional determinant.

4.3 3.3 Partial-Wave Decomposition

Exploiting $O(4)$ symmetry, expand the Green function in hyperspherical harmonics:

$$G_{ij}(x, x') = \sum_{\ell} G_{ij}^{(\ell)}(r, r') \mathcal{Y}_{\ell}(\Omega, \Omega'), \quad (18)$$

where $\mathcal{Y}_{\ell}$ are the appropriate angular eigenfunctions, and $G_{ij}^{(\ell)}(r, r')$ satisfy coupled radial Green-function equations. This reduces the 4D problem to a tower of 1D radial problems for each partial wave ℓ .

5 4. One-Loop 1PI Effective Action and Determinant

5.1 4.1 One-Loop Effective Action

The one-loop correction to the effective action is formally

$$\Gamma^{(1)} = \frac{1}{2} \ln \frac{\det \mathcal{M}}{\det \mathcal{M}_{\text{FV}}}, \quad (19)$$

where $\mathcal{M}_{\text{FV}}$ is the fluctuation operator evaluated in the false vacuum. In terms of Green functions this can be written as a trace over fluctuations, e.g. via a heat-kernel or spectral representation:

$$\ln \det \mathcal{M} = \text{Tr} \ln \mathcal{M}. \quad (20)$$

5.2 4.2 1PI Effective Action and Renormalisation

The full 1PI effective action can be written schematically as

$$\Gamma[\phi_1, \phi_2] = S_E[\phi_1, \phi_2] + \Gamma^{(1)}[\phi_1, \phi_2] + \cdots, \quad (21)$$

where $\Gamma^{(1)}$ is expressible in terms of the fluctuation determinant. Ultraviolet divergences manifest in $\Gamma^{(1)}$ and are removed by introducing counterterms in the bare action:

$$\mathcal{L} \rightarrow \mathcal{L} + \delta\mathcal{L}, \quad (22)$$

with $\delta\mathcal{L}$ containing mass, coupling, and field-strength counterterms. The renormalisation conditions are chosen (e.g. in $\overline{\text{MS}}$ or on-shell schemes) to match physical parameters (masses, couplings) in a specified renormalisation scheme.

In the Green-function method, renormalisation is implemented by subtracting from the fluctuation contributions the corresponding terms expanded around the false vacuum and, if necessary, additional local counterterms to cancel remaining divergences.

6 5. Tunnelling Rate with One-Loop Corrections

Combining the classical bounce action and the one-loop corrections, the decay rate per unit volume takes the schematic form

$$\frac{\Gamma}{V} = A_{\text{1-loop}} \exp [-B - \Gamma^{(1)}], \quad (23)$$

where B is the classical bounce action and $\Gamma^{(1)}$ encodes the one-loop determinant ratio between the bounce background and the false vacuum. The prefactor $A_{\text{1-loop}}$ also includes contributions from zero modes (e.g. translational zero modes) and possible negative modes, as well as overall normalisation factors.

In the two-field case, the structure of the calculation is the same as in the one-field Green-function treatment, but the fluctuation operator and Green functions are matrix-valued, and the renormalisation conditions must be specified for the multi-field potential.

7 Conclusion

This roadmap summarises the main steps required to compute the false-vacuum decay rate in a theory with two interacting scalar fields using the Green-function method at one loop:

1. Identify all minima and select the false and true vacua.
2. Solve the coupled bounce equations and compute the classical action difference B .
3. Construct the quadratic fluctuation operator and the associated Green functions.
4. Evaluate the one-loop effective action via the determinant ratio, implementing renormalisation.
5. Combine these ingredients to obtain the tunnelling rate per unit volume, Γ/V .

This mirrors the known single-field formalism while capturing the full multi-field structure relevant for realistic models.